

NumPy and Pandas in Python: Complete Guide

1. NumPy (Numerical Python)

Definition

NumPy is a powerful Python library used for numerical computations. It provides support for large, multi-dimensional arrays and matrices, along with a collection of mathematical functions to operate on these arrays.

Core Concepts

The central object in NumPy is the ndarray (N-dimensional array). It allows fast operations and efficient storage compared to Python lists.

Key concepts include:

- Homogeneous data types
- Vectorization
- Broadcasting
- Memory efficiency

Conceptual Implementation

NumPy works by storing data in contiguous memory blocks, enabling fast computation. Operations are applied element-wise without explicit loops.

Common Functions and Their Purposes

1. `array()` → Create arrays
2. `arange()` → Generate range of numbers
3. `zeros()` / `ones()` → Create arrays filled with zeros or ones
4. `reshape()` → Change array shape
5. `mean()`, `median()`, `std()` → Statistical functions
6. `dot()` → Matrix multiplication
7. `sqrt()`, `exp()`, `log()` → Mathematical operations
8. `sort()` → Sort elements
9. `unique()` → Find unique values
10. `where()` → Conditional selection

Advantages

- Faster than Python lists
- Memory efficient
- Supports complex mathematical operations
- Foundation for many data science libraries

Use Cases

- Scientific computing
- Machine learning
- Data preprocessing
- Image processing

2. Pandas

Definition

Pandas is a data analysis and manipulation library built on top of NumPy. It provides data structures like Series and DataFrame for handling structured data.

Core Concepts

Key data structures:

- Series: 1D labeled array
- DataFrame: 2D labeled table
- Indexing and alignment
- Handling missing data

Conceptual Implementation

Pandas builds on NumPy arrays and adds labels (rows and columns). It allows intuitive data manipulation similar to spreadsheets and SQL.

Common Functions and Their Purposes

1. `read_csv()` → Load data
2. `head()`, `tail()` → View data
3. `info()`, `describe()` → Summary
4. `loc[]`, `iloc[]` → Data selection
5. `drop()` → Remove rows/columns
6. `fillna()` → Handle missing values
7. `groupby()` → Group data
8. `merge()` → Combine datasets
9. `sort_values()` → Sort data
10. `apply()` → Apply functions

Advantages

- Easy data manipulation
- Handles missing data
- Integrates with NumPy
- Powerful for data analysis

Use Cases

- Data cleaning
- Data analysis
- Financial analysis
- Machine learning preprocessing

3. NumPy vs Pandas

NumPy is used for numerical operations on arrays, while Pandas is used for structured data handling.

NumPy is faster for mathematical operations, whereas Pandas is better for labeled data and analysis.